COOLING ENVIRONMENTS

Maximizing Cooling Performance in Data Centers by Optimizing Fluid Velocity Limits

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VELOCITY LIMITATIONS IN LIQUID COOLING

- **Are today's fluid velocity limits leaving cooling performance and rack density on the table?**
- Current standards for velocity limits restrict flow to **~3-6 ft/s (1-2 m/s)** and are presumably driven by issues commonly encountered in residential and facility piping systems
- Other fluid power industries utilize **much higher** velocities through rubber hoses; **some above 30 ft/s (9 m/s)** which indicates improvements in hose velocities up to an order of magnitude for the data center industry
- **Why it matters? More flow = more cooling potential = potentially higher rack density**

DATA CENTER HOSE

- Data center hoses are highly engineered with specific system reviews to avoid constraints caused by components and internal geometry features
- Data centers contain relatively little valving in systems which prevents large changes in velocity from constrained openings or abrupt stoppages of flow

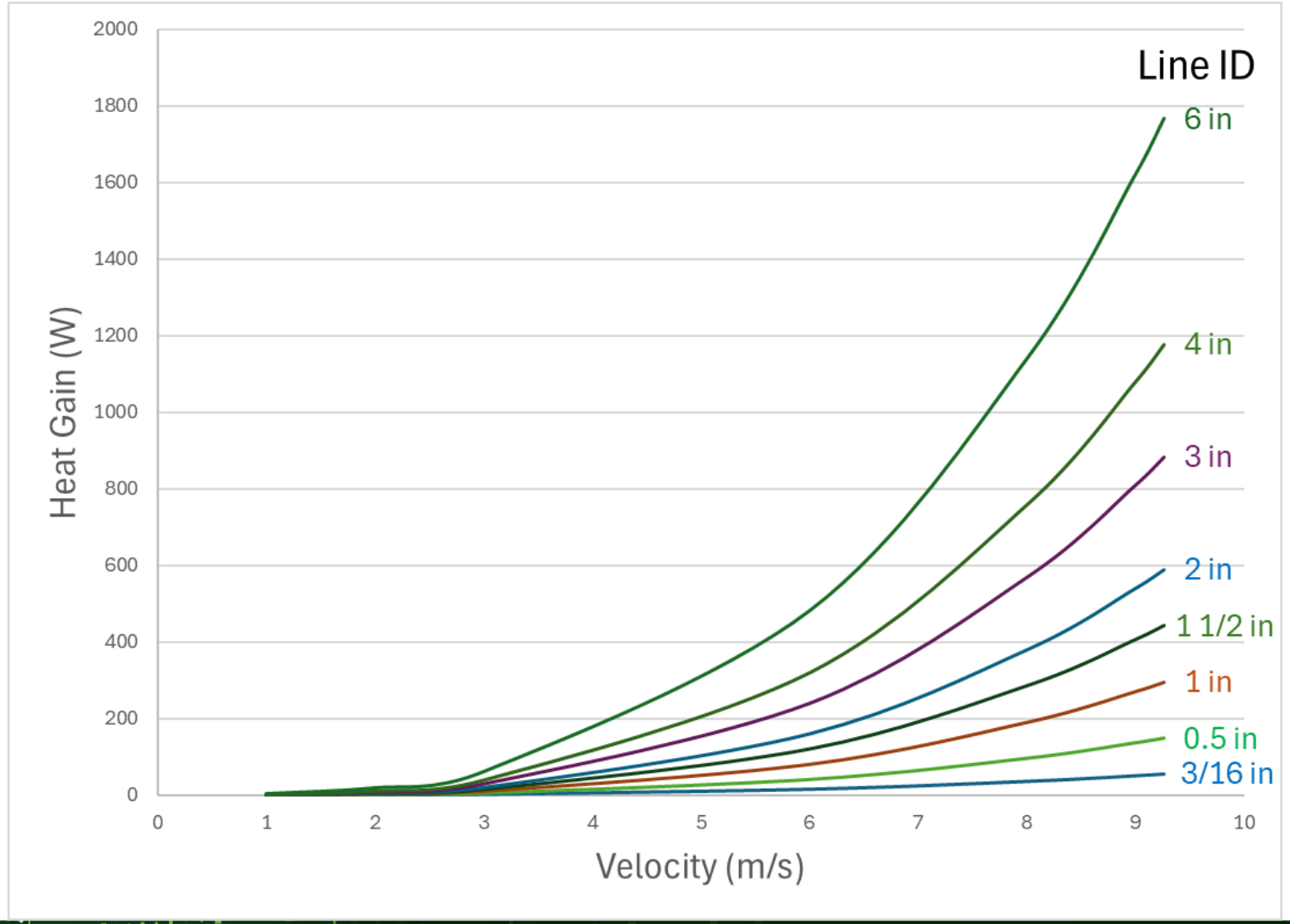
- Rubber Hoses:
  - Provide compliance for easy routing and connections in systems
  - Provide dampening in conditions of pressure variations, component vibration, and noise
  - Minimize heat transfer through walls
  - Durability because of separation of reinforcement layer from fluid and environmentally contacting layers (Slurry example)

WHAT HAPPENS WHEN WE INCREASE VELOCITY?

- Mass flow rate primarily determines available heat transfer capacity
  - $V = \dot{m}/(\rho \times A)$
- Turbulent flow is not bad as long as it does not compromise heat transfer capacity
- Do not want to add more heat than collecting or rejecting
- Pressure drop is proportional to the heat gain; higher velocities can increase both
- Cavitation is a concern where high velocity differentials exist at boundaries
- Wall shear can develop due to a high rate of flow at fluid conductor edges and could represent a long-term material degradation risk
- **The goal isn't maximum velocity, the goal is optimal velocity**

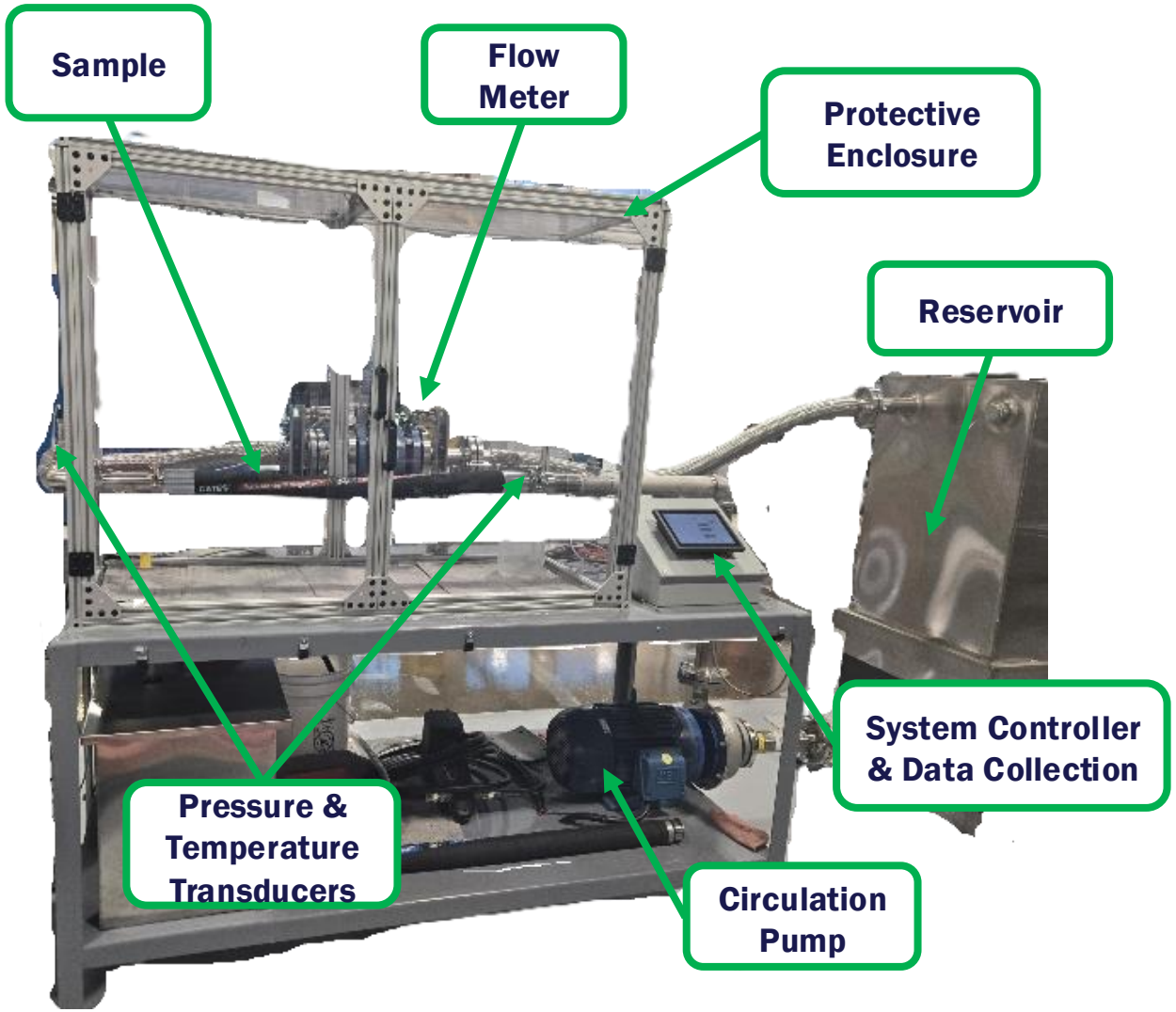
ANALYSIS OF MECHANICS

- Increasing velocity accelerates heat gain in fluid
- Effects like turbulence are secondary
- Effects at system boundaries minor (such as cavitation, wall shear, etc.)



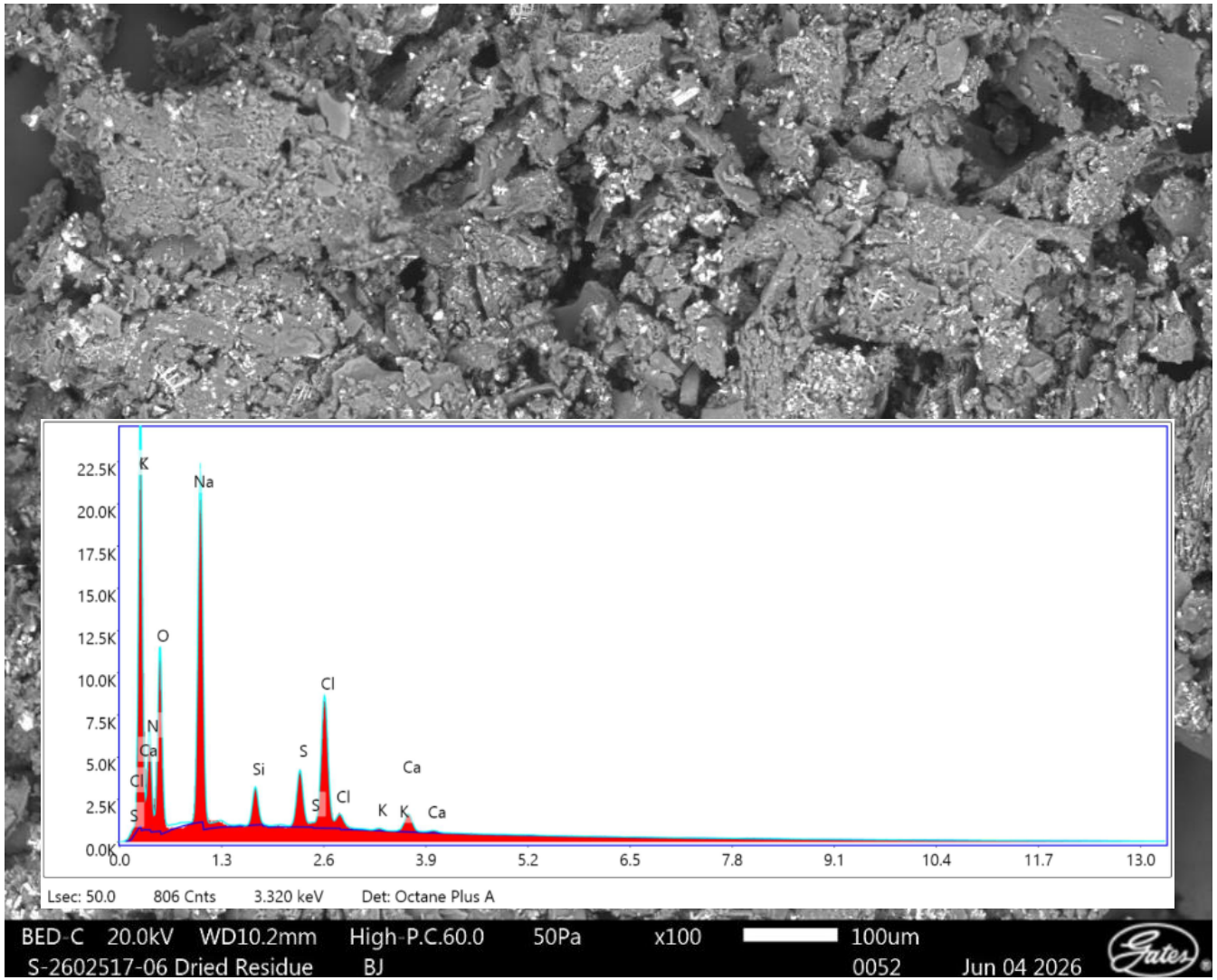
WHAT WE'RE TESTING: EMPIRICAL VALIDATION

- Test set up to allow direct validation of the impact of flow rates on performance
- Data collection: System Transducers, Fluid Properties, Representative Materials, Hose Scanning
- Artificial aging conditions introduced to predict long-term effects



Fluid Test Results

- Polyglycol content
- Solution pH
- Chemical Analysis/FTIR
- Fluid Viscosity
- Corrosion Inhibitor Concentration



CONCLUSION

The Velocity Limit Is Inherited, Not Tested – Until Now

Today's ceiling **~1–2 m/s** came from piping practice, not liquid cooling

Other fluid power industries run **above 9 m/s** through hose

The goal isn't maximum velocity. It's **optimal** velocity

More Flow → More Cooling → Higher Rack Density

RESOURCES

- Fluid Mechanics, Frank M. White
- Uniform Plumbing Code (1957) ; BMS66 (1940)
- The Importance of Velocity Limits in a Piping System
- Womack's Fluid Power Data Book - Amazon
- SPEED LIMITS: Relating flow velocity to hydraulic performance | Machine Design
- Piping Handbook (Nayaar)
- Gates Product Application
- Gates Fluid Flow Calculator
- Gates Website

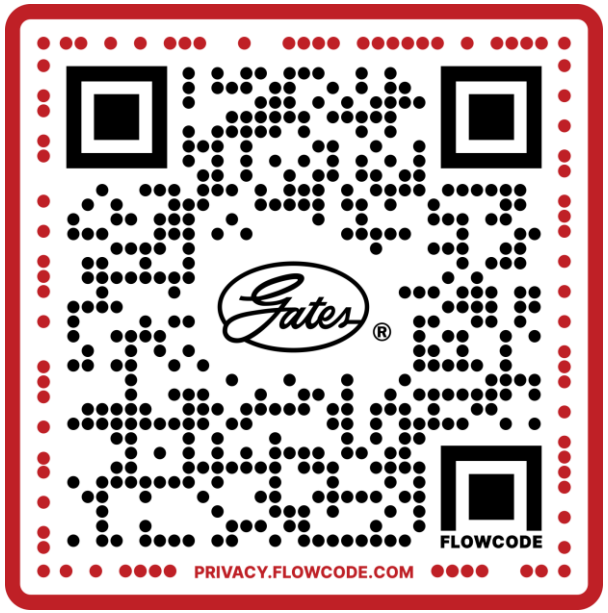
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CALL TO ACTION:

- **Help us set the next flow velocity guidelines!**
  - Suggestions for modifications or results you'd like to see.
  - Share similar results, especially non-rubber materials, for comparative analysis
  - Seeking collaborators for a flow rate optimization spec or sub-project.

- Where to find additional information
  - Data Center Cooling Solutions
  - Scan the QR Code or visit the Gates Corporation in Hall 3, Booth #D32 to connect with us, speak with an expert, and learn more!



Thank You!

